

Comparison of different airfoils in ground effect across various angles of attack

Samuel Cech | Charlie Petersen | Rhys Shields

Background:

When a downforce producing wing is run close to the ground it experiences what is called “Ground Effect” which accelerates flow between the airfoil and the ground, increasing the magnitude of the low-pressure zone. Our presentation modeled ground effect on airfoils in 2D to quantify the flow behavior. Using CFD is an effective way to quickly iterate through the effects that different parameter changes have on an airfoil's aerodynamic forces.

Project Goals:

Our primary goal was to simplify the process of iterating through design changes on a downforce producing airfoil in ground effect. Our code is designed to automatically generate a mesh for any airfoil data file across any value of height from ground, angle of attack, and chord length. This allows you to set the conditions you want to iterate through and then simply call the necessary airfoil data file when calling your code, no need to manually remake your mesh with every change in condition.

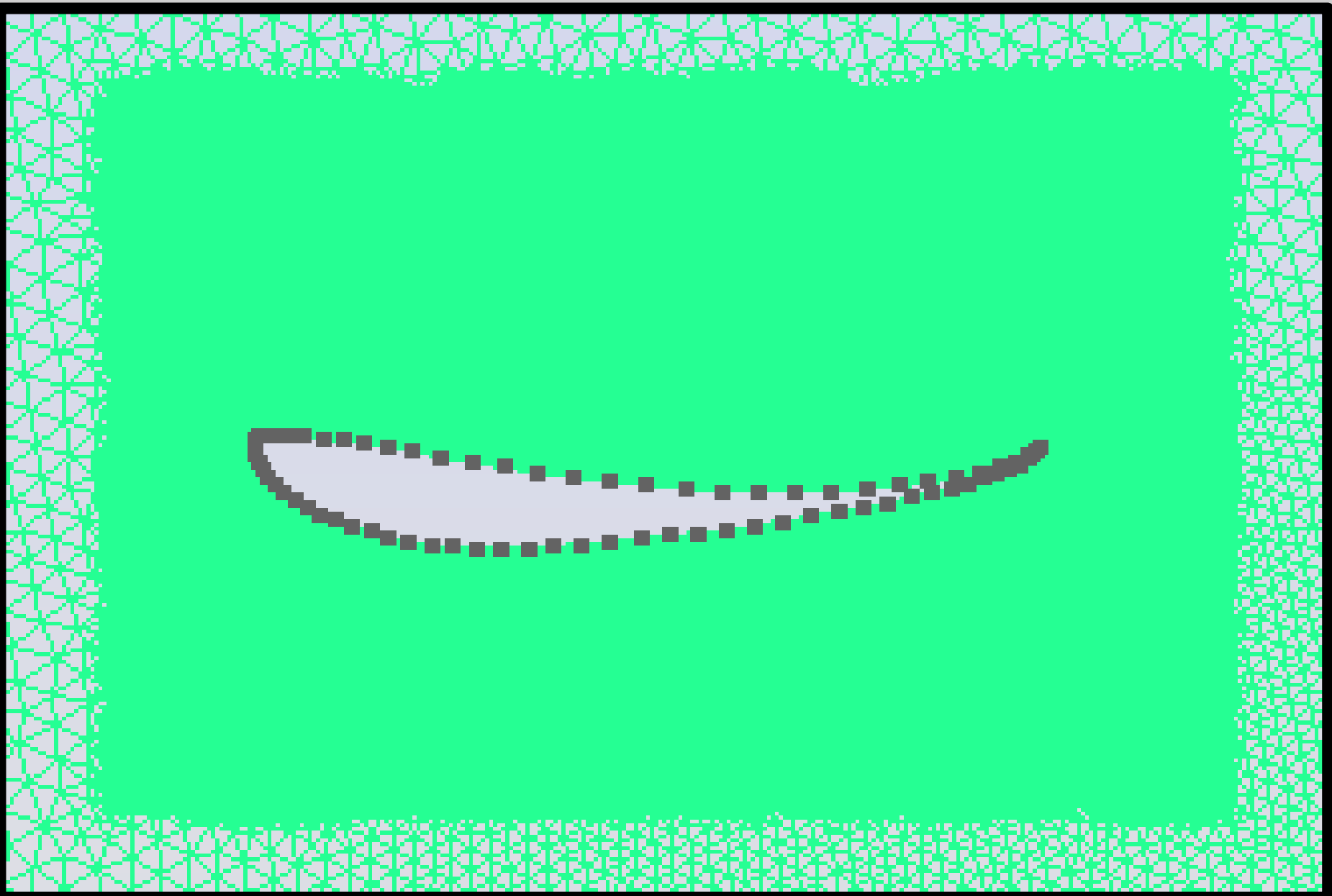
Method:

For our simulations we choose a NACA 2412 with a chord of 1 meter, 1 meter from the ground, at -8, 0, and 8 degrees AOA in the same FEniCS environment. The simulation operates at accurate sea level conditions with a high Reynolds number of roughly 838,000. The Smagorinsky model with Large Eddy Meshing (LES) was used to better simulate turbulent flow.

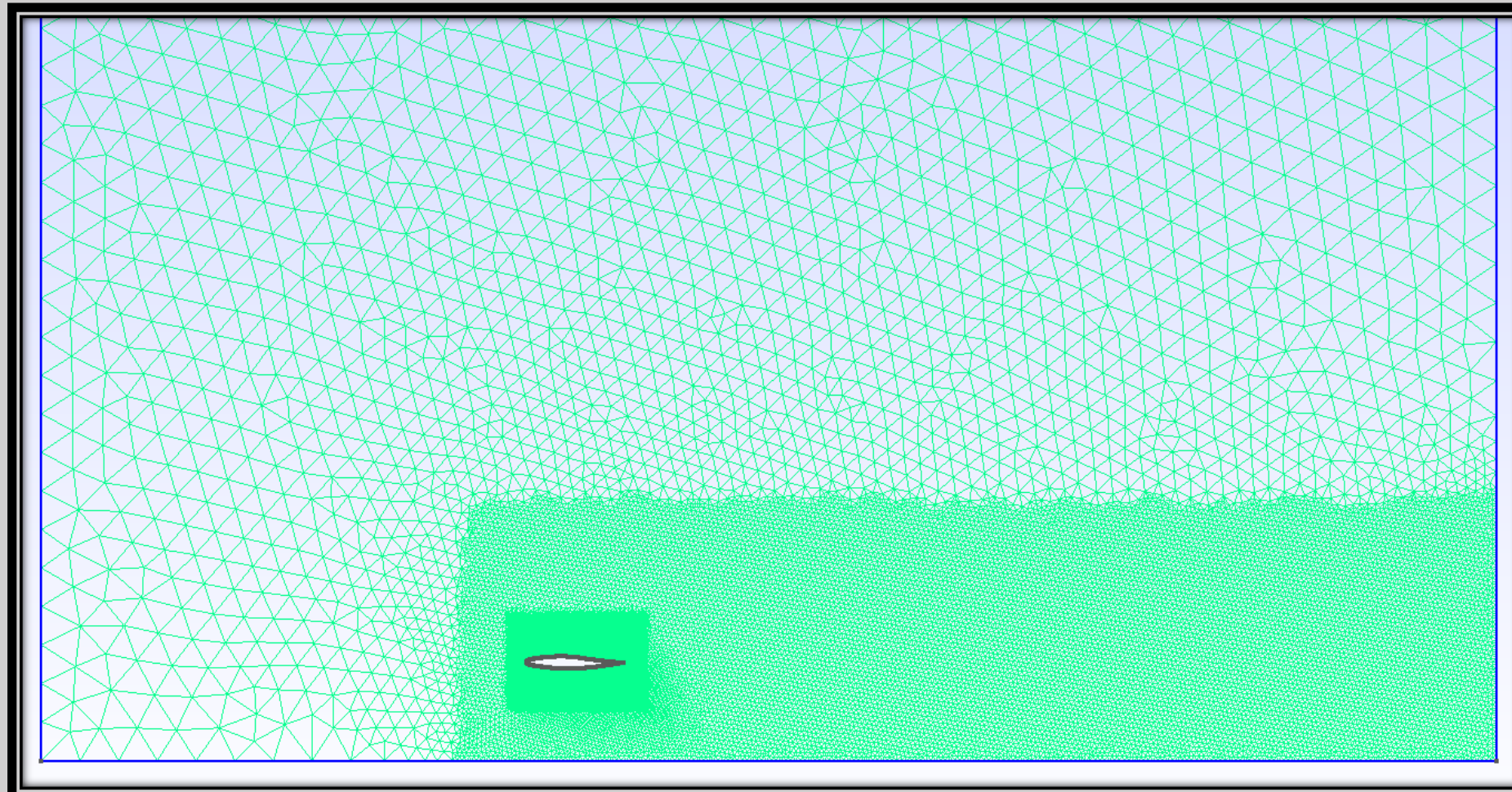
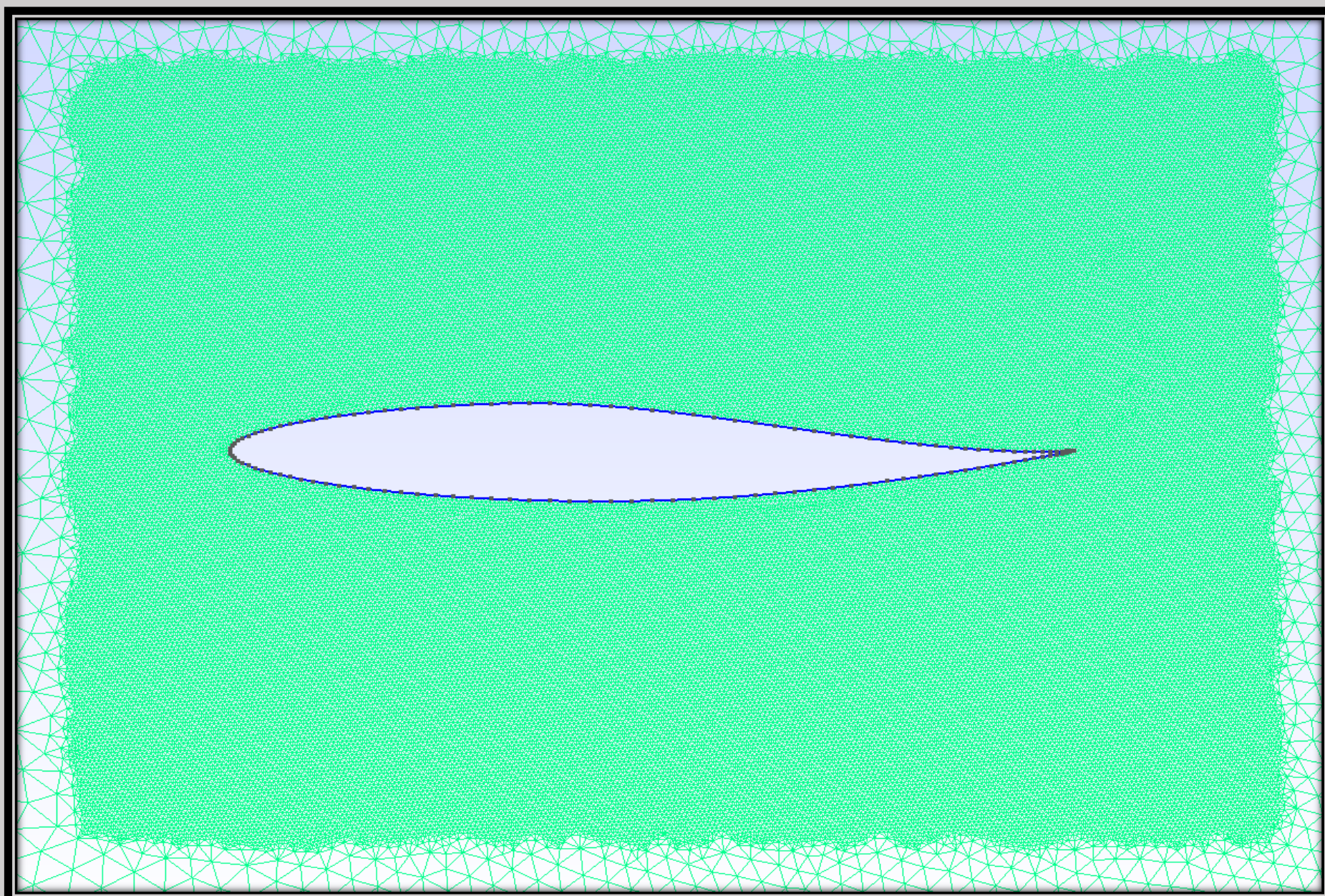
15 m/s flow NACA 2412

Mesh Possibilities

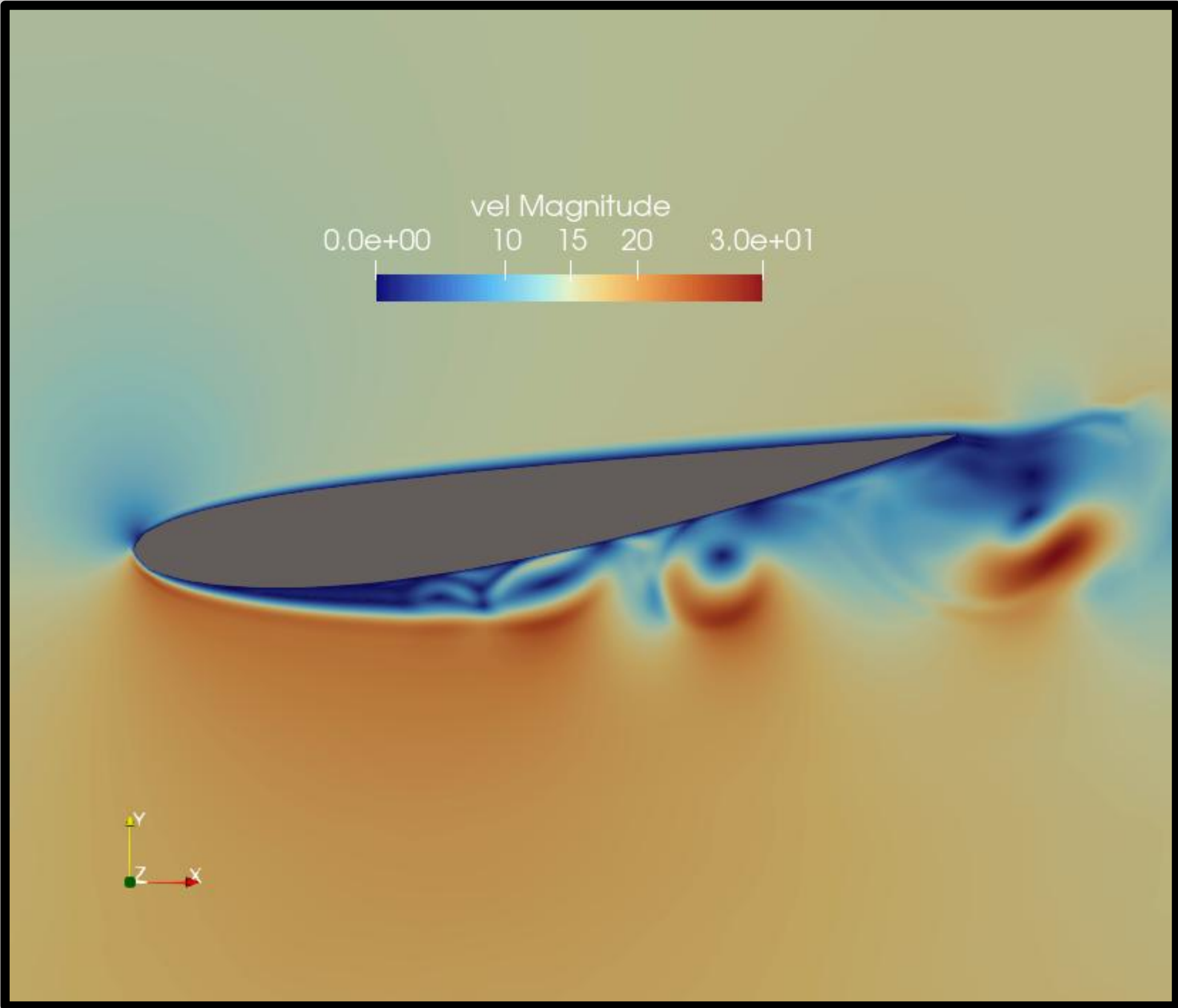
Selig S1223



RAE 2822

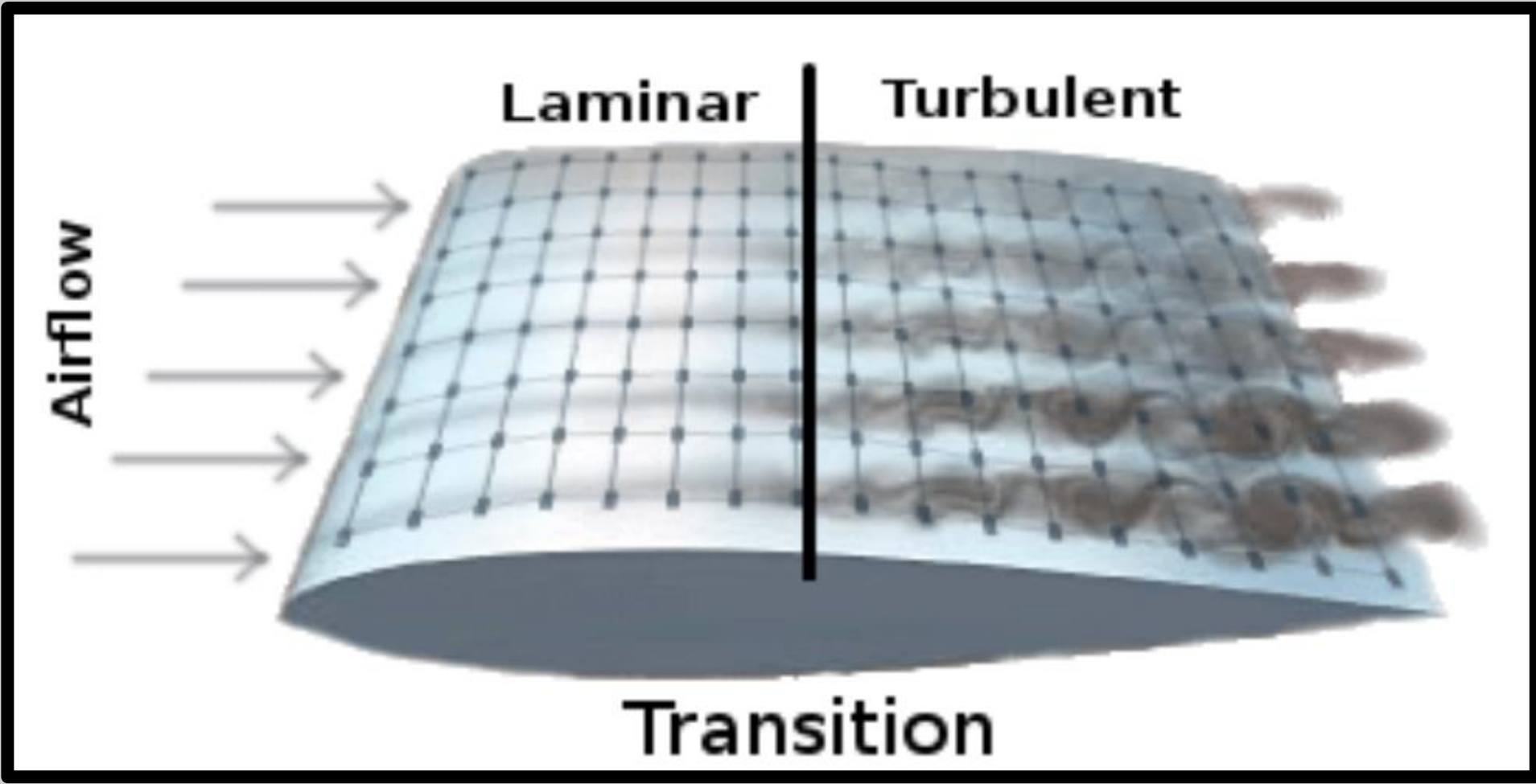


8° Rotation

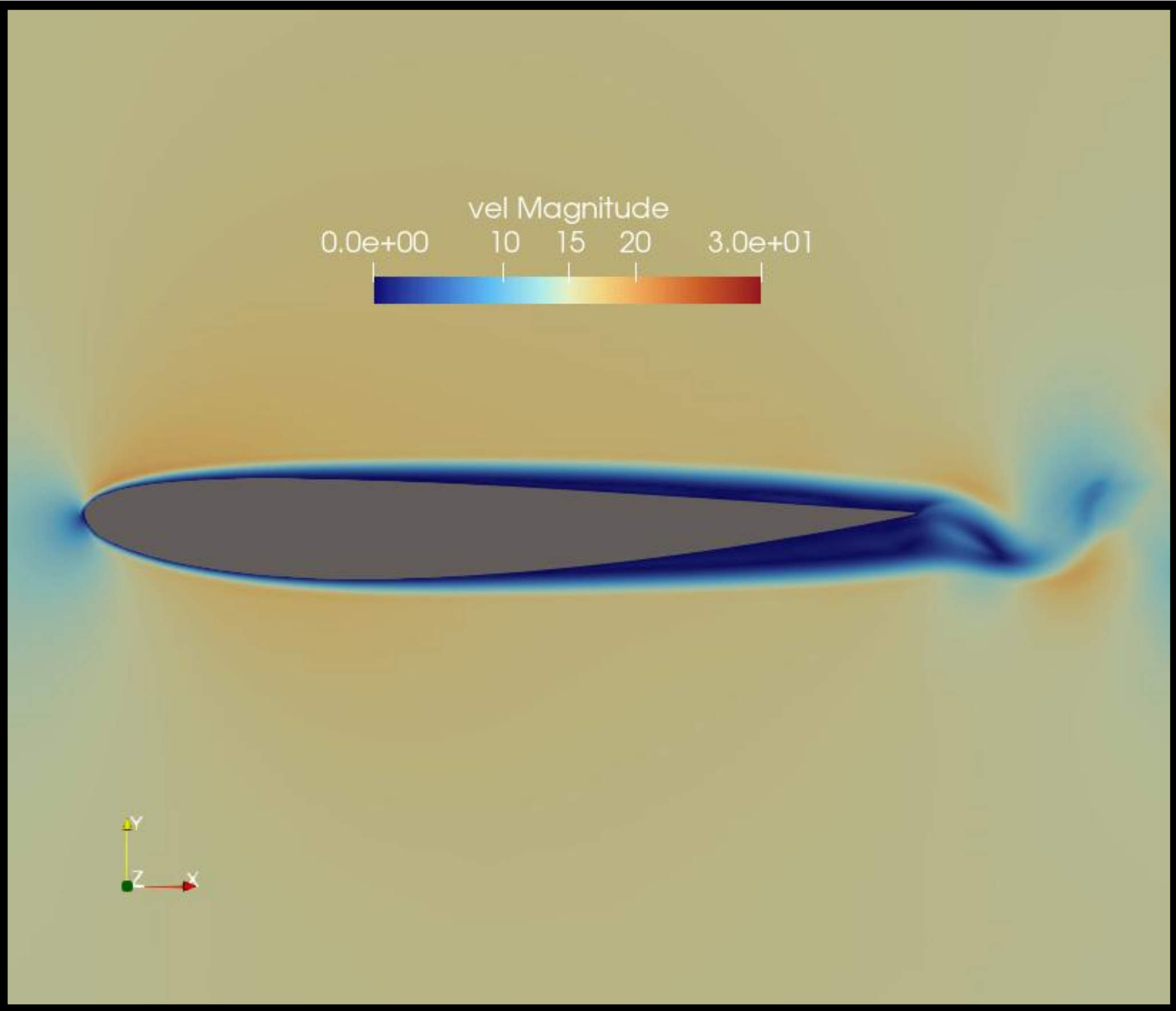


Observations

- Earlier transition to turbulence than expected in $\pm 8^\circ$ sims
- Large boundary layer on 0-degree airfoil
- In $\pm 8^\circ$ sims, the side facing the airflow remains fully laminar.



0° Rotation



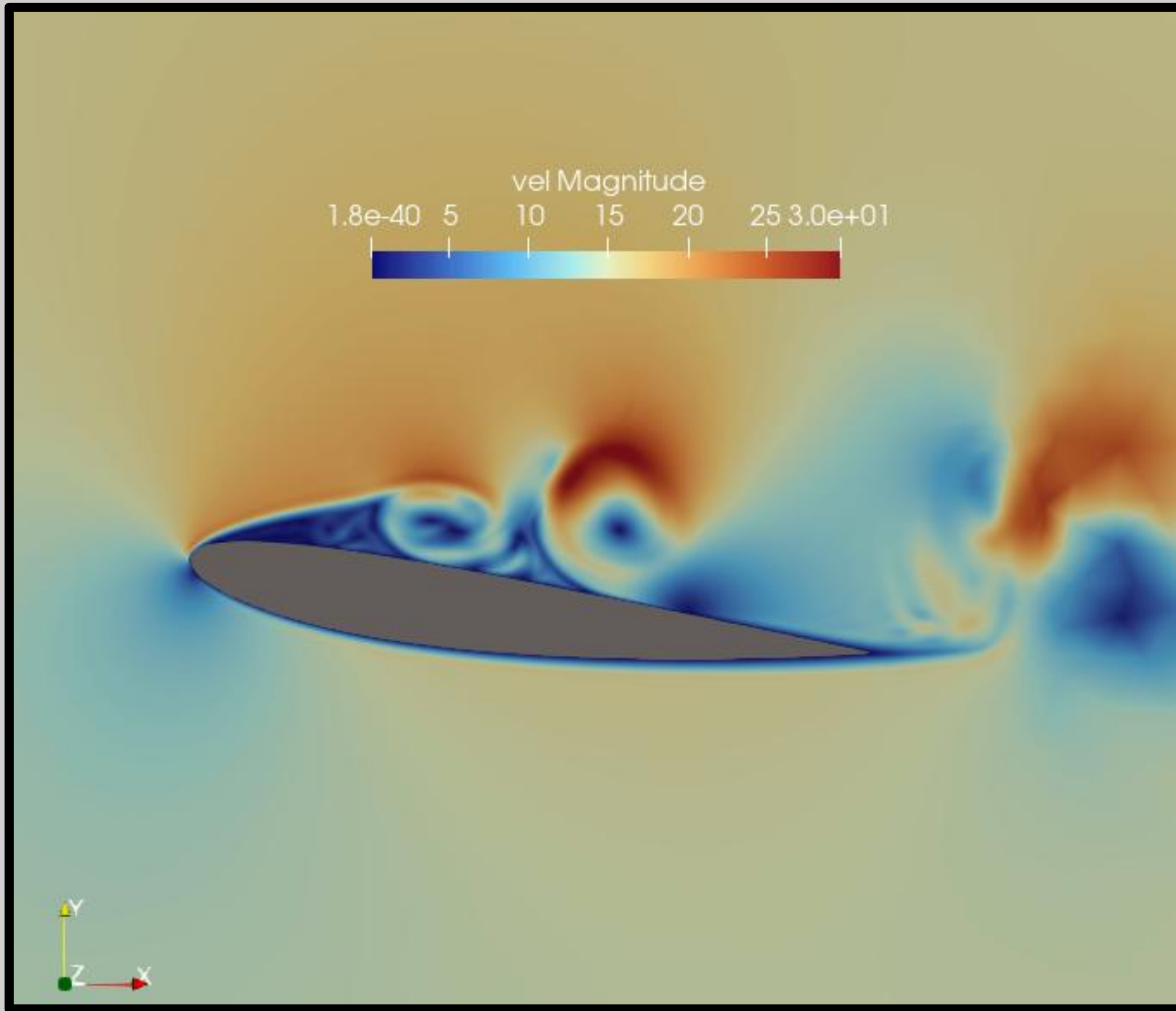
Script Parameters

- Free Stream Velocity: 15.0 m/s
- Fluid Density: 1.0 kg/m³
- Fluid Dynamic Viscosity: 1.789e-5 Pa
- Reynolds number: 838,000

Future Work

- Improve turbulence modeling
- Compare more airfoil geometries
- Refine mesh near airfoil and in the wake
- Expand study across more angles of attack and other parameters
- Validate results with published experimental data

-8° Rotation



Meshing Parameters

- Boundary: 9x15
- Elements: 75000
- Nodes: 37000
- ADEQUATE for LES with Smagorinsky

